

Predicting Speed-Dating Match Outcomes Using Multilayer Perceptrons and Dimensionality Reduction

Mini-Project 1: COGS 209 Scientific Data Analysis and Statistical Learning

Lead Author: Travis Ramirez

Co-Authors: Bryan Zhu, Dylan Walmsley, Jiaguan Pan

Student Group: Meow

Date: 05/10/2026

1 Introduction and Hypotheses

Human social interactions involve rapid evaluations of compatibility integrating large amounts of heterogeneous information, including demographic characteristics, self-perceptions, interests, and interpersonal preferences. Matching markets (dating apps or events) try to pair people using stated preferences and profile attributes, yet behavioral work often finds mismatches between *what people say they want* and *what predicts their actual choices* when meeting candidates. Speed dating provides a useful experimental framework for studying these decisions because participants make repeated, real-time judgments under relatively controlled conditions. Because these decisions likely emerge from complex nonlinear interactions between multiple variables, machine learning approaches may provide useful tools for modeling the hidden structure underlying match outcomes. Traditional linear methods may fail to capture these interactions, whereas neural networks can learn distributed feature representations capable of modeling more complex decision boundaries.

In this project, **we investigated whether multilayer perceptron (MLP) models could successfully predict match outcomes in a speed dating dataset and how architectural choices influence model generalization.** Specifically, we examined the effects of hidden layer dimensionality, dropout-based regularization, and principal component analysis (PCA)-based feature reduction on classification performance. Because the dataset is moderately high-dimensional and class imbalanced, we investigated how model complexity influences overfitting and whether lower-dimensional representations improve robustness. We hypothesized that **(1)** smaller MLP architectures (fewer hidden units) would generalize better due to reduced risk of overfitting, **(2)** higher dropout rates would improve validation performance by acting as a stronger regularizer, and **(3)** PCA-based dimensionality reduction would further improve generalization by removing redundant and noisy features, leading to better test accuracy and ROC-AUC compared to models trained on the full feature set.

2 Methods

2.a Description of Dataset

This dataset was found under the name [SpeedDating](#) on [OpenML.org](#) (dataset link can be found in the Reference Section below). This data was collected from participants in experimental speed dating events between 2002 and 2004. During these events, the attendees engaged in a four-minute "first date" with every other participant of the opposite sex. At the end of their four minutes, participants were asked if they would like to see their date again. The dataset also includes questionnaire data gathered from participants at different points in the process. However, because each participant interacted with multiple individuals during the events, **a single participant may appear across multiple observations in the dataset. This violates the assumption that all observations are statistically independent.** For example, some participants may consistently rate all of their partners more positively than others, causing multiple rows associated with that participant to share similar patterns. If treated as fully independent observations, the model would incorrectly interpret these repeated tendencies as stronger evidence than they actually are, potentially inflating statistical significance or model performance.

The primary goal of analyzing this dataset revolves around understanding the factors that influence dating decisions and the formation of matches in a speed dating context. This dataset contains information capturing various attributes of participants and their interaction. It is composed of 8,378 rows and initially 123 columns. **Each row** represents **one speed dating interaction** that either results or does not result in a match. The dataset **columns** include a rich variety of information describing the interaction, such as: **Demographic Information-**

age, gender, race, and field of study for each participant, **Preferences**- participants' stated preferences for various attributes (e.g., attractiveness, sincerity, intelligence) in a potential partner, **Self-Perceptions**- how participants rate themselves on these same attributes, **Interests**- ratings of various leisure activities (e.g., sports, movies, reading) and **Dating Decisions**- whether a participant decided to 'match' with their speed date partner, and whether the partner also decided to match, leading to an actual match. The data types included 57 columns of *float64*, 7 columns of *int64*, and 59 columns of *object*. Columns with *object* data types were examined to identify **nominal categorical variables** suitable for one hot encoding. This was done by inspecting unique values and their counts for each object column. The categorical features identified, specifically 'gender', 'race', 'race_o', and 'field', were transformed using one-hot encoding. This approach was chosen to convert these nominal categories into a numerical format suitable for machine learning algorithms, avoiding any false sense of ordinality.

2.b Preprocessing: Missing Values and Categorical Variables

Before proceeding with imputation, a **comprehensive check for null values, empty strings, duplicate rows, and zero values was performed** to understand the extent of data quality issues. We set a **threshold of 30% missing data**, such that variables with more than 30% missing values were excluded from the analysis. The column *'expected_num_interested_in_me'* was dropped from the dataset due to a very high percentage of missing values (approximately 78.5%). Retaining this column would have introduced significant noise and bias, making it unsuitable for reliable imputation or direct use (**Table 1**). The data also contained **56 columns prefixed with 'd_'**, which largely represented binned **string values of derived numerical values**. These *'d_ columns'* were typically summaries of other columns, for example, *'d_age'* redescrbing the original age of the 25 year old subject was represented in the bin *'25-30'*. It was decided to drop all 56 *'d_ columns'*, as they contained redundant binned string representations of summary data that were less informative after one-hot encoding. For the remaining numerical columns with missing values, **median imputation** was chosen. **This method was preferred over mean imputation** because it is more robust to outliers, ensuring that extreme values do not influence the imputed values and maintain the overall distribution of the features.

Examination of the distribution of *'field'*, which represents participants' academic backgrounds, revealed **substantial imbalance and high granularity across categories**. In **Figure 1**, some fields such as *"field_stats"* contained very few observations (6 subjects), whereas others such as *"field_law"* contained hundreds (604 subjects). This imbalance made the variable overly sparse and less effective for statistical modeling. To address this issue, related **academic fields were grouped into broader categories using a custom group_field function** that classified entries according to discipline-related keywords (e.g., "STEM," "Business," and "Law"). These broader categories (**Figure 2**) were then converted into numerical features using one-hot encoding. Grouping reduced the cardinality of the variable, improved statistical reliability for sparsely represented categories, and produced a more interpretable representation for downstream analyses such as PCA and machine learning models.

Percentage of Missing Values per Column:

	Missing Percentage
expected_num_interested_in_me	78.515159
expected_num_matches	14.000955
shared_interests_o	12.843161
shared_interests_partner	12.735736
ambitious_o	8.617809
ambition_partner	8.498448
met	4.476009
funny_o	4.296968
funny_partner	4.177608
guess_prob_liked	3.688231
intelligence_o	3.652423
intelligence_partner	3.533063
sinsere_o	3.425639
sincere_partner	3.306278

Table 1: Percentage of missing values across all dataset features

Distribution of Top 5 and Least 5 Academic Fields

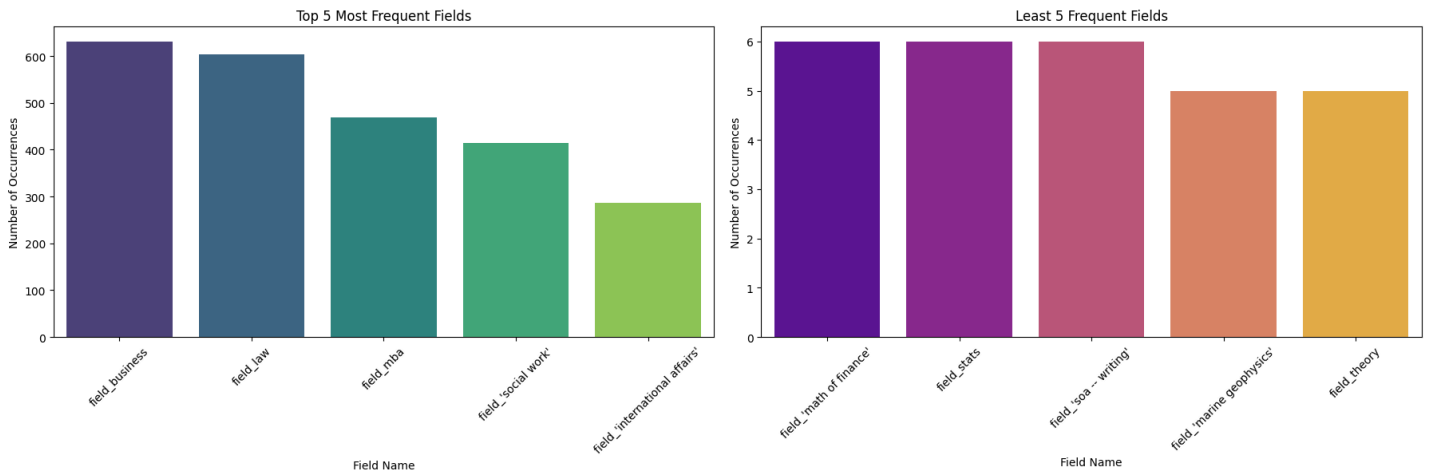


Figure 1: Distribution of Top 5 Academic Fields *Before* Grouping

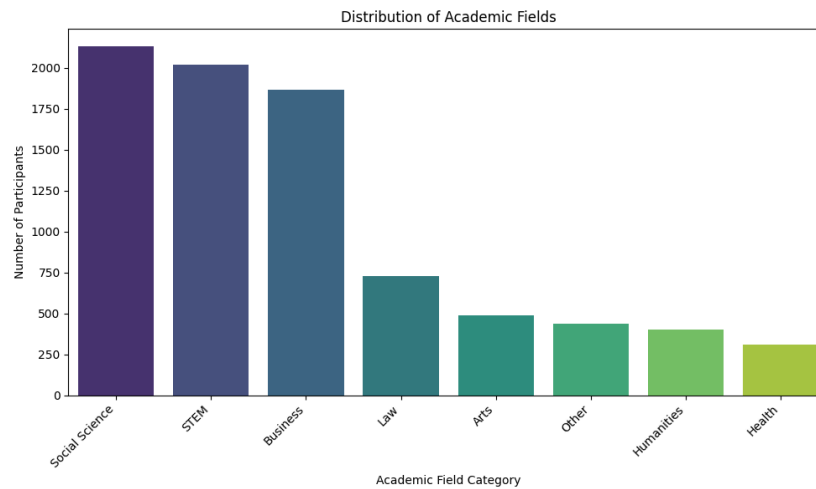


Figure 2: Distribution of Academic Fields *After* Grouping

2.c Exploratory Data Analysis

As part of the exploratory data analysis, distributions were examined for all numerical variables to identify potential anomalies, inconsistencies, and differences in scale across features. Histograms and summary statistics revealed substantial variation in the ranges of numerical values within the dataset. In **Figure 3**, some variables such as age were represented as larger integer-valued quantities, whereas many questionnaire-based variables were measured on bounded rating scales ranging from 1–10. Because these **variables exist on different numerical scales**, features with larger magnitudes could disproportionately influence downstream analyses and machine learning models. To address these differences in scale, **numerical variables were standardized using z-score normalization**, such that each feature was transformed to have a mean of 0 and a variance of 1. Standardization is particularly important for distance-based and gradient-sensitive machine learning methods, where variables with larger scales can dominate distance calculations. Standardization also improves numerical stability and comparability across features during model training.

having little practical value for identifying successful matches. Because of this imbalance, evaluation metrics beyond test accuracy, such as F1 score and ROC-AUC, are necessary to provide a more balanced assessment of model performance, particularly with respect to the minority match class.

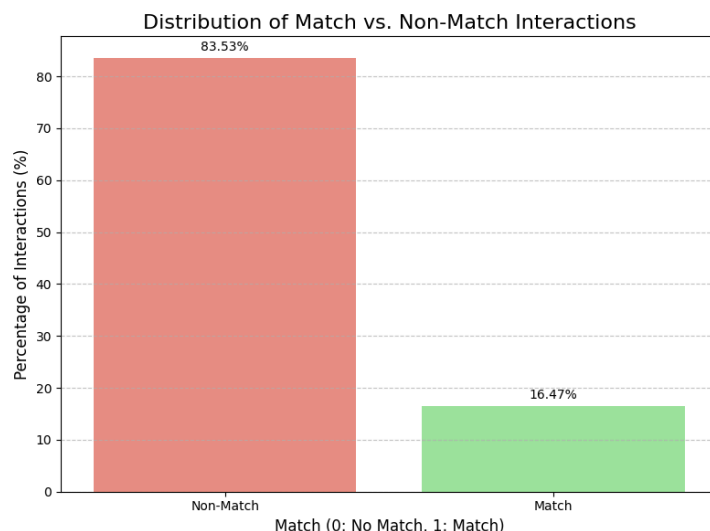


Figure 4: Distribution of Match vs Non-Match Interactions

2.d Classification Models

The classification model of choice that was chosen to predict matches between observations was that of the **Multi-Layer Perceptron (MLP)**. A standard MLP architecture of two hidden weight layers followed by ReLU activations and a dropout layer to facilitate regularization was used.

The **dimensions of the hidden layers** and **degree of dropout** were hyperparameters that we tuned during our experiments. In particular, input vectors were first fed to a hidden layer that expanded vector representations from input dimension $x_0 \rightarrow h_0$, where h_0 is the dimension of the first hidden layer. After a standard ReLU and dropout layer, another hidden layer projects from $h_0 \rightarrow h_1 = h_0/2$ (i.e. reduces hidden representation to dimension $h_0/2$) and ReLU $\rightarrow$ Dropout. Finally, an output layer projects $h_1 \rightarrow 1$, which is thresholded to be either 0 or 1, thus yielding a prediction for whether a given interaction is a match or not.

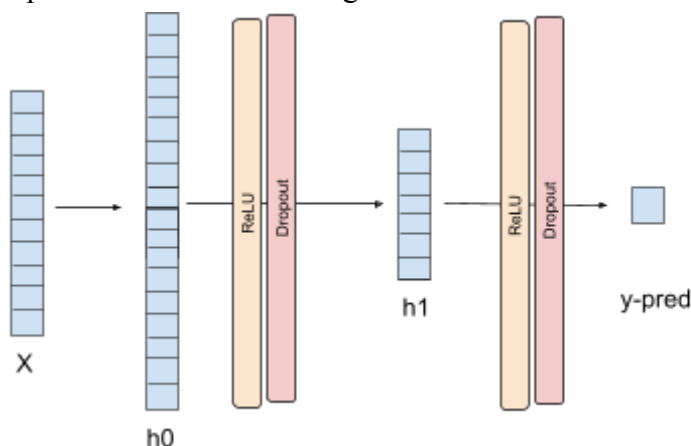


Figure 5: Visualization of MLP Architecture

To thoroughly evaluate the performance of a variety of MLP constructions based on varying hyperparameters, a **grid search based set of experiments was performed** where a new MLP model was retrained each time with a **different of h_0 and dropout degree**, varying from [8, 16, 32, 64, 128] and [0.3, 0.5] respectively. The best model parameters would be chosen based on the accuracy on the test set, which is withheld from the start.

The **reasoning behind choosing a MLP is two-fold**. **1)** Given the Universal Approximation Theorem, we hypothesized that a MLP of sufficient complexity should be able to approximate the underlying “function” that determines match compatibility assuming sufficient data. **2)** While the internal weights and parameters of a MLP are not entirely interpretable, the architecture and hyperparameters that we ultimately choose can be a benchmark of the complexity of the problem—i.e. a larger number of hidden units necessary for good performance can be an indicator of problem complexity.

2.e Feature Reduction Technique

The raw data is moderately high dimensional; for one observation, the raw data contains 123 features. During preprocessing, we dropped redundant or uninformative columns to reach 55 variables. While this is not an excessive amount, we wanted to test our hypothesis that a reduced feature set could improve performance.

We chose to use **Principal Component Analysis** as a feature reduction technique. In this technique, the features are chosen such that components capturing some degree of the variance of the data are chosen, where all components are orthogonal to each other. We choose this technique because we **hypothesize that certain features matter more when determining partner matches and, therefore**, contribute more to the overall variance within the dataset. The PCA processor was set up to choose principal components that capture 85% of the variance, ultimately yielding 34 features. It is upon this dataset that we re-run the same set of MLP’s and evaluate the performance.

An important consideration when performing PCA was to ensure that the testing data was properly withheld while fitting the PCA model to **avoid leakage across training and testing data** and avoid artificially inflated test accuracy. Thus, our **preprocessing pipeline was developed such that the test data is withheld before PCA**.

3 Results - Model Comparison and Visualization

3.a Experiment 1 Results: MLP without feature reduction

We first evaluated the performance of an MLP on our data without feature reduction. 10 experiments were run (based on a set of 5 hidden layer dimensions versus a set of 2 dropout degrees). Training and validation loss were tracked for each set of parameters, and the results for the baseline model are shown in **Figure 6**. Briefly, during model training, the more complex models perform better, with 128 units in the MLP performing the best. Moreover, 0.3 dropout outperforms 0.5 dropout within the same level of model complexity. The results reverse on the validation set, where less complex models (8-16 units) perform much better than the more complex models. This reversal of performance between the training and validation sets indicates that the more complex models overfit to the training data, memorizing data, and generalize to new data very poorly. In testing multiple levels of model complexity/flexibility, we found a balance within 8-16 units.

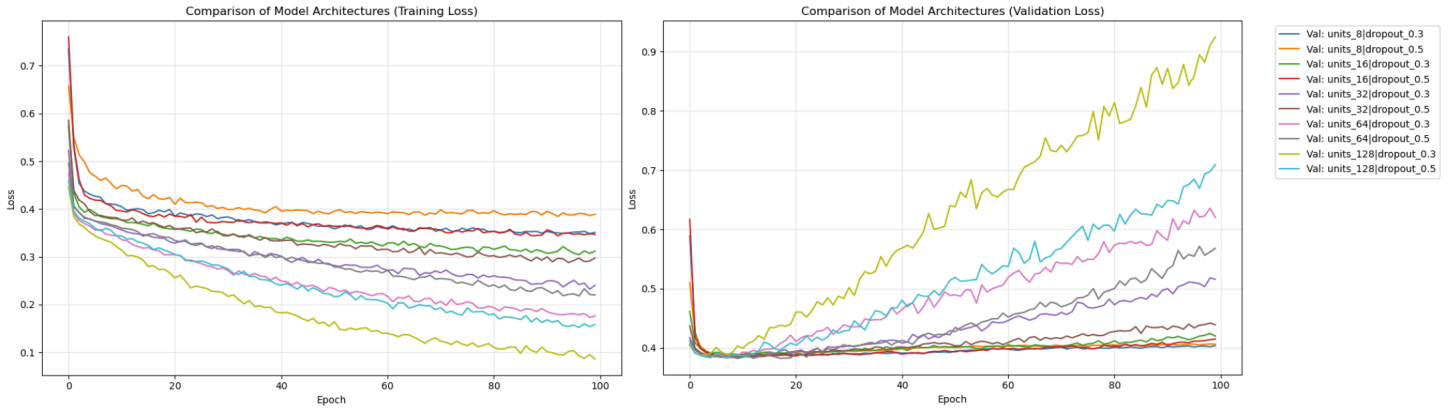


Figure 6: Training Loss (Left) and Validation Loss (Right) per Epoch for Hyperparameter Configurations (Varying Hidden Units and Dropout Rate)

3.b Experiment 2 Results: MLP after PCA

A second set of experiments was performed after conducting PCA, yielding dimension-reduced data. Once again, training and validation loss was tracked for each set of parameters, and the results for the models trained on PCA dimension-reduced data are as follows:

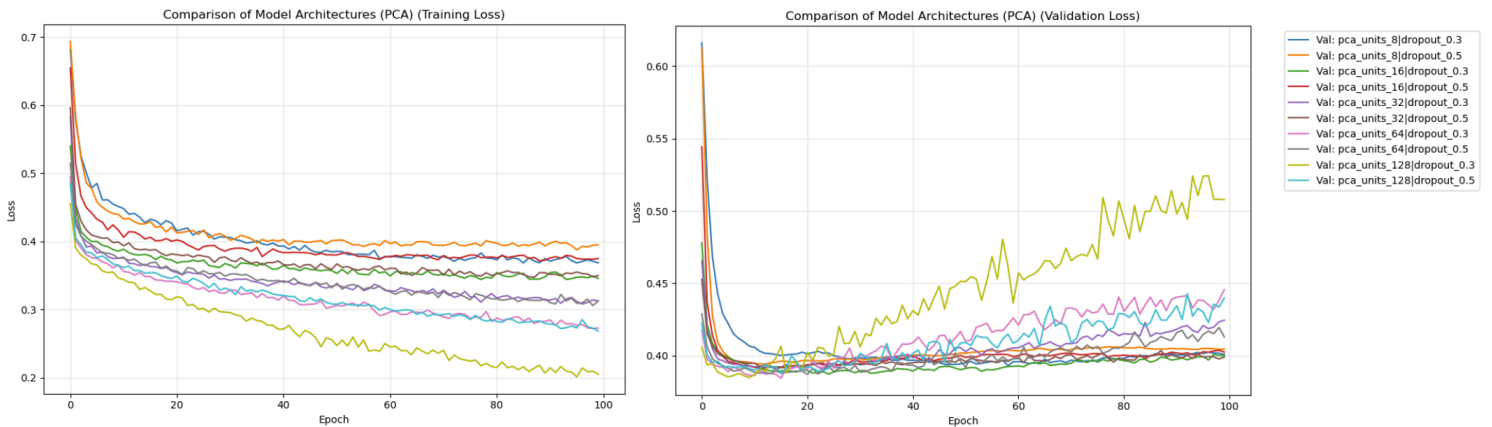


Figure 7: Training Loss (Left) and Validation Loss (Right) per Epoch for Hyperparameter Configurations (Varying Hidden Units and Dropout Rate)

3.c Experimental Findings

We can observe some patterns with regards to the training process of the MLP's based on their training and validation loss graphs that are consistent across Baseline-MLP's and PCA-MLP's. Firstly, when observing the effect that an increased hidden dimension (i.e number of hidden units) has on the models, it is evident that **with more units we can observe steeper drops in training loss. However, validation loss tends to increase sharply with increased units.** This inverse relationship between training and validation loss indicates that larger models are significantly more prone to overfitting. While a low number of units performs quite well (8 to 32 units), anything with or more than 64 units shows increases in testing loss from overfitting. This suggests that to model this relationship, only a few units are necessary while the other units effectively serve to memorize the training data and fail to generalize to unseen data.

Another pattern observed was that **higher dropout rates tended to produce better regularization performance**. When comparing a pair of models that share the same number of units, we can see that the model trained with dropout rate 0.5 tends to achieve a lower validation loss than the model trained with dropout rate 0.3. This observation suggests that the dropout further aims in mitigating overfitting and improving generalization on unseen data.

Beyond training dynamics, we also **evaluated each model on F1 score and ROC-AUC**, as test accuracy alone is misleading given the class imbalance in the dataset — roughly 83% of observations are non-matches, meaning a model can achieve high accuracy simply by predicting ‘no match’ every time. Several configurations reported an F1 score of 0.0, indicating the model defaulted to predicting the majority class entirely. The best F1 score achieved was only ~0.32. ROC-AUC proved more informative, ranging from 0.74 to 0.77, with the best baseline result from the 16-unit / dropout=0.5 model at 0.772.

The test accuracy for the ten parameter sets of MLP models for Baseline and on PCA data are summarized in the table below.

Model	Baseline Accuracy	PCA Accuracy
units_8 / dropout_0.3	0.8353	0.8359
units_8 / dropout_0.5	0.8353	0.8353
units_16 / dropout_0.3	0.8341	0.8401
units_16 / dropout_0.5	0.8353	0.8353
units_32 / dropout_0.3	0.8383	0.8401
units_32 / dropout_0.5	0.8353	0.8419
units_64 / dropout_0.3	0.8305	0.8359
units_64 / dropout_0.5	0.8359	0.8473
units_128 / dropout_0.3	0.8282	0.8365
units_128 / dropout_0.5	0.8335	0.8401

Table 2: Test accuracy of MLP models on baseline and PCA-transformed data. Test accuracy of MLP models on baseline and PCA-transformed data. Each model varied in the number of hidden units (8–128) and dropout rate (0.3 or 0.5). Overall, PCA preprocessing produced modest improvements in classification accuracy for most parameter settings, with the highest performance observed for the model with 64 hidden units and 0.5 dropout (PCA accuracy = 0.8473).

4 Discussion

The central aims of this project were to evaluate whether multi-layer perceptron architectures could learn the complex, non-linear structure of human matching decisions, and whether dimensionality reduction would improve MLP performance. Our results offer a nuanced picture: while MLPs did learn meaningful structure from the data, their generalization was substantially limited by overfitting and class imbalance, two challenges that weigh on the validity of conclusions drawn about what predicts human match outcomes.

Consistent with our first hypothesis, smaller architectures generalized better. The validation loss curves revealed that larger hidden layer configurations severely overfit the data, and validation loss diverged sharply from

training loss well before epoch 100. Small networks, on the other hand, maintained relatively stable validation performance. This pattern may suggest that in similar datasets, more complex MLPs are prone to learning noise rather than generalizable structure in the match decision process. In other words, the aforementioned mismatch between stated and revealed preferences may introduce irreducible noise that more flexible models memorize instead of learning latent structures.

Our second hypothesis—that higher dropout rates would improve generalization—was partially supported. Dropouts of 0.5 consistently produced less overfitting than the 0.3 dropout rate. However, dropout rate alone was unable to rescue generalization inefficiencies, demonstrating that regularization cannot fully compensate for excessive model complexity relative to dataset size.

While PCA yielded marginal improvements across configurations, the gains were modest, contrary to our third hypothesis. This is likely because MLP networks can implicitly learn feature selection during training. As a result, the explicit dimensionality reduction from PCA offers limited additional benefit.

A key limitation of the current models is the class imbalance in the dataset. The near-zero F1 scores observed in several configurations suggest that the models defaulted to predicting the majority class. This error is consequential because our goal is to understand the contributions to the minority outcome: a mutual match. Future work could address this through techniques such as class weighting or oversampling strategies (e.g. SMOTE) to improve minority class recall. Additionally, many models appeared to reach their optimal validation loss well before Epoch 99, suggesting that early stopping could be beneficial in future experiments to prevent unnecessary overfitting and reduce training time.

Other improvements to future work could include early stopping, preventing the unnecessary overfitting observed in models that reach optimal validation loss well before epoch 100. Another limitation of the current approach is that one-hot encoding treats all categorical groups as equally distinct, which may not capture semantic similarities between related academic fields. Future work could explore learned continuous embeddings that represent categorical variables in lower-dimensional spaces while preserving relationships between categories. However, such approaches generally require larger and more balanced datasets to produce reliable representations.

Altogether, these findings suggest that MLPs and PCA are promising strategies for modeling the hidden structure of human matching decisions. The behavioral complexity that makes speed dating an interesting test case may require not just better models, but richer, larger, and more balanced datasets to yield reliable insights.

Reference

1. Raymond Fisman, Sheena S. Iyengar, Emir Kamenica, Itamar Simonson, Gender Differences in Mate Selection: Evidence From a Speed Dating Experiment, *The Quarterly Journal of Economics*, Volume 121, Issue 2, May 2006, Pages 673–697, <https://doi.org/10.1162/qjec.2006.121.2.673>

Dataset

<https://www.openml.org/search?type=data&sort=runs&id=40536&status=active>

Github Repository

https://github.com/ZhuZiLiBryan/COGS209_Speed_Dating_Project